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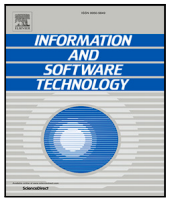

Understanding and evaluating software reuse costs and benefits from industrial cases—A systematic literature review

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ABSTRACT

Context: Software reuse costs and benefits have been investigated in several primary studies, which have been aggregated in multiple secondary studies as well. However, existing secondary studies on software reuse have not critically appraised the evidence in primary studies. Moreover, there has been relatively less focus on how software reuse costs and benefits were measured in the primary studies, and the aggregated evidence focuses more on software reuse benefits than reuse costs.
Objective: This study aims to cover the gaps mentioned in the context above by synthesizing and critically appraising the evidence reported on software reuse costs and benefits from industrial cases.
Method: We used a systematic literature review (SLR) to conduct this study. The results of this SLR are based on a final set of 30 primary studies.
Results: We identified nine software reuse benefits and six software reuse costs, in which better quality and improved productivity were investigated the most. The primary studies mostly used defect-based and development time-based metrics to measure reuse benefits and costs. Regarding the reuse practices, the results show that software product lines, verbatim reuse, and systematic reuse were the top investigated ones, contributing to more reuse benefits. The quality assessment of the primary studies showed that most of them are either of low (20%) or moderate (67%) quality.
Conclusion: Based on the number and quality of the studies, we conclude that the strength of evidence for better quality and improved productivity as reuse benefits is high. There is a need to conduct more high quality studies to investigate, not only other reuse costs and benefits, but also how relatively new reuse-related practices, such as InnerSource and microservices architecture, impact software reuse.

1. Introduction

Software reuse helps companies improve product quality [1–6] and development productivity [2–5]. Studies also reported that software reuse helps faster time-to-market [7,8] and better performance [3,9]. Software reuse has benefits but also comes with additional costs. In particular, developing reusable assets may incur additional costs, such as asset design [10] and team coordination [11]. However, these additional costs pay off when the reusable assets are reused in multiple products and solutions [5,11].

Several approaches have been used to implement software reuse, such as software product lines [10,12,13], reuse from repository [4,14,15] and component reuse [16]. In addition, depending on whether companies have a well-thought-out strategy to develop for and with reuse, software reuse can be classified into systematic, ad hoc, and controlled reuse [10]. According to Tomer et al. [10], systematic reuse refers to the scenario wherein companies have planned to mine and catalog the reusable assets into a shared repository and design them for later reuse. On the other hand, ad hoc reuse (also referred to as opportunistic reuse) represents the cases wherein companies do not have a shared repository for reusable assets, and teams need to look for them and use them independently. Lastly, companies may have a shared repository for reusable assets but have yet to design them for later reuse, referred to as controlled reuse. Studies also investigated different types of reuse, namely verbatim reuse (also referred to as black-box reuse or reuse as-is) and modified reuse (also referred to as white-box reuse).

Primary studies, such as Succi et al. [1] and Morisio et al. [2], concluded software reuse benefits from empirical evidence. Such primary studies were further aggregated and synthesized in secondary studies (e.g., literature reviews or mapping studies) to draw new conclusions or identify patterns on software reuse costs and benefits. Mohagheghi and Conradi [17] performed a systematic literature review (SLR) on industrial quantitative studies investigating the effects of software reuse on quality, productivity, and economics. Barros et al. [20]

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Table 1
Secondary studies related to software reuse costs and benefits.

Related work (RW)	Year	Software reuse costs and benefits focus	Gap (G)
RW1[17]	2007	Impacts in quality, productivity, and return on investment	G1: the year of the included primary studies is up till 2005. G2: the study lacks quality assessment on primary studies. G3: the study did not discuss the relationship between reuse-specific characteristics and software reuse costs/benefits.
RW2[18]	2015	Mainly benefits and few costs	G2: the study lacks quality assessment on primary studies. G3: the study did not discuss the relationship between reuse-specific characteristics and software reuse costs/benefits. G4: not all the included primary studies include an industrial case. G5: the study did not provide the metrics used to measure software reuse costs/benefits.
RW3[19]	2016	Reusability influence on non-functional requirements	G2: the study lacks quality assessment on primary studies. G3: the study did not discuss the relationship between reuse-specific characteristics and software reuse costs/benefits. G4: not all the included primary studies include an industrial case. G5: the study did not provide the metrics used to measure software reuse costs/benefits.
RW4[20]	2018	Only benefits	G2: the study lacks quality assessment on primary studies. G4: not all the included primary studies include an industrial case. G5: the study did not provide the metrics used to measure software reuse costs/benefits.

conducted a systematic mapping study (SMS) investigating what software reuse benefits have been brought to the industry. Bombonatti et al. [19] conducted an SMS investigating the relationship between non-functional requirements and software reusability. Younoussi and Roudies [18] conducted an SLR on software reuse with one of their research questions on its benefits.

The aforementioned systematic secondary studies are mainly about software reuse benefits, with a minimal focus on software reuse costs. Furthermore, except Mohagheghi and Conradi [17], none of them [18–20] report how their included primary studies measured the identified software reuse benefits. Likewise, except for Mohagheghi and Conradi [17], their results are not based on industrial cases only, such as reporting practitioners' general opinions on the usefulness of software reuse practices or studies based on an analysis of some open-source systems. Moreover, Mohagheghi and Conradi [17] included the primary studies up to 2005 in their review published in 2007, which indicates the need for updating their review. Among the four identified systematic secondary studies on software reuse costs and benefits, only Barros et al. [20] investigated the relationship between different reuse practices and reuse benefits, e.g., which reuse methods (systematic reuse and ad hoc reuse) may lead to more reuse benefits. Lastly, none of the existing systematic secondary studies on software reuse costs and benefits performed a rigorous quality assessment on their included primary studies, e.g., assessing the quality of reporting (see details in Section 2). Without a rigorous quality assessment, it is unclear how strong the primary evidence behind the software reuse costs and benefits results reported in these systematic studies. We aim to address these gaps by identifying and appraising the evidence on software reuse costs and benefits reported in the empirical studies based on industrial cases only. To do that, we conducted an SLR with the following specific contributions:

- Identified the software reuse costs and benefits from industrial cases.
- Identified the context and reuse-specific characteristics of the industrial cases reported in the included studies and associated them with software reuse costs and benefits.
- Identified the metrics used in the primary studies to measure software reuse costs and benefits.
- Assessed the quality of the primary study to gauge evidence for software cost and benefit results.
- Identified research gaps in the existing literature and suggested future research directions.

The remainder of the paper is structured as follows: Section 2 presents the related work. Section 3 describes the research methods and the validity threats. Section 4, Section 5, Section 6, and Section 7 answer the four research questions, and Section 8 discusses the findings. Section 9 concludes the SLR and proposes future work.

2. Related work

Many studies have reported impacts when conducting software reuse, such as performance decay [21], better product quality [1,7,9, 22], and improved productivity [2,16,23,24].

We identified seven secondary studies that are directly related to [17–20] or mentioned [25–27] software reuse costs and benefits. Table 1 summarizes the characteristics of four directly related secondary studies [17–20] together with their corresponding research gaps (note: Gx means Gap no. x). The identified gaps indicate a need for a new secondary study on software reuse costs and benefits. All gaps are discussed in Section 1. Here, we further elaborate on the quality assessment gap (G2). Mohagheghi and Conradi [17] did not assess the quality of the primary studies but discussed the reported validity threats. Younoussi and Roudies [18] performed a quality assessment based on the research questions to check if the primary study contains the needed information. Bombonatti et al. [19] performed quality assessments on primary studies using bibliometric impact information. However, they

lack aspects related to content quality since the authors only consider the venue rank and citation number. Barros et al. [20] had a research question on validity threats, but the authors only checked whether the validity threats were reported. Thus, it is hard to understand how reliable the evidence is reported in the primary studies.

The main objectives of the other three secondary studies are not on software reuse costs and benefits. However, they mentioned software reuse costs and benefits in their results. Chen and Babar [25] mentioned software reuse benefits when introducing one primary study about the effect of the variability management approach. Neto et al. [26] mentioned software reuse benefits when answering the research question about the testing strategies adopted in software product line testing approaches. As for Flemstrom et al. [27], they mentioned vertical test reuse benefits when answering the research question about the justification or motivation for using vertical test reuse.

Our study aims to investigate the state of the art of reported software reuse costs and benefits from industrial cases published until 2021. We want to identify how software reuse costs and benefits were measured in the primary studies and understand which reuse practices benefit companies more. In addition, we plan to assess and analyze the strength of the evidence behind the software reuse costs and benefits claims in the included primary studies.

3. Research methodology

We followed the systematic literature review guidelines proposed by Kitechenham et al. [28] to conduct our study.

3.1. Research questions

We addressed the following research questions (RQs) in this study:
RQ1: What are the characteristics of the primary studies on software reuse costs and benefits?
RQ2: What are the software reuse costs and benefits reported in the primary studies, and how are they related to reuse practices?
RQ3: How are software reuse costs and benefits measured in the primary studies?
RQ4: What is the strength of reported evidence on software reuse costs and benefits in the primary studies?

The characteristics help refine the understanding of how companies practiced software reuse in the included primary studies. The characteristics include general information about the included primary studies (e.g., year and venue), contextual characteristics (e.g., development methodology), and reuse-specific characteristics (e.g., reuse types).

3.2. Search and study selection processes

We used a combination of automated and snowballing search strategies. Snowballing search strategy needs a good start set [29], which we identified from the automated search strategy. To identify the primary studies on software reuse costs and benefits using automated search, we tried Scopus using the keywords "software reus*, cost, benefit" and retrieved 3482 primary studies. It was challenging to manage such a large number of primary studies. Therefore, we decided to follow an indirect approach — extract the primary studies that secondary studies included to synthesize software reuse costs and benefits. To make our search more inclusive, we first used automated search to identify secondary studies on software reuse. And within the identified secondary studies on software reuse, we included the ones about software reuse costs and benefits. Then, we extracted all relevant primary studies on software reuse costs and benefits from the included secondary studies. We followed the snowballing guidelines from Wohlin [29] to identify more primary studies on software reuse costs and benefits. To conclude, we performed the search in three phases (see Fig. 1).

- Phase 1: Identification of secondary studies using automated search.
- Phase 2: Identification of the primary studies from the secondary studies identified in Phase 1.
- Phase 3: Forward snowballing on the primary studies identified in Phase 2.

3.2.1. Phase 1: Identification of secondary studies using automated search

In Phase 1, we aimed to identify all secondary studies on software reuse. We identified the search string keywords - "software reuse", "literature review", and "mapping study" and ran the automated search in four databases[1]: Scopus, IEEE, Web of Science, and ACM. We conducted the search string execution in May 2022 and limited the year to 2021. After removing all the duplicates, we retrieved 427 secondary studies on software reuse.

To avoid missing relevant secondary studies on software reuse and validate the reliability of the automated search results, we included two tertiary studies on software reuse [30,31] as an additional data source. Ultimately, we had 481 secondary studies on software reuse.

3.2.2. Phase 2: Identification of the primary studies from the secondary studies identified in Phase 1

Phase 2 consists of two steps — Step 1: identify secondary studies on software reuse costs and benefits from the identified studies in Phase 1, and Step 2: identify primary studies on software reuse costs and benefits from the identified studies in Step 1.

In Step 1, after all authors had a shared understanding of Secondary Study Selection Criteria (see Section 3.3), we piloted the criteria on the title and abstract of ten randomly selected secondary studies, resulting in no disagreements. Subsequently, the first author independently applied the criteria to all secondary studies and included 79 secondary studies and seven doubtful secondary studies. The other two authors reviewed the selection results, including 81 secondary studies on software reuse. The first author independently read each research question (RQ) from the included secondary studies and identified 13 relevant secondary studies on software reuse costs and benefits. All authors independently read the identified 13 secondary studies. In a joint meeting, we discussed the RQs and agreed to include seven secondary studies on software reuse costs and benefits [17–20,25–27], which four [17–20] are directly on software reuse costs and benefits and three [25–27] have at least one RQ about software reuse costs and benefits.

In Step 2, we extracted 146 relevant primary studies, which were used to answer RQs about software reuse costs and benefits from the identified seven secondary studies in Step 1 after removing the duplicates. We developed Primary Study Selection Criteria (see Section 3.3) to include primary studies investigating software reuse costs and benefits in industrial cases and reached a shared understanding of the criteria through a joint discussion. The first and second authors piloted Primary Study Selection Criteria on ten randomly selected primary studies, finding no disagreements in the pilot results. The first author read the full text and independently applied Primary Study Selection Criteria to all candidate primary studies, including 29 primary studies and eight doubtful primary studies. The second author reviewed the selection results, and ultimately, we included 29 primary studies on software reuse costs and benefits.

3.2.3. Phase 3: Forward snowballing on the secondary and primary studies identified in Phase 2

In Phase 3, we conducted snowballing on the included four secondary studies directly on software reuse costs and benefits, and 29 included primary studies from Phase 2.

Since the primary studies are extracted from the secondary studies, which went through a thorough search strategy, we applied forward

[1] The detailed search string for each database is available at https://zenodo.org/records/10649264.

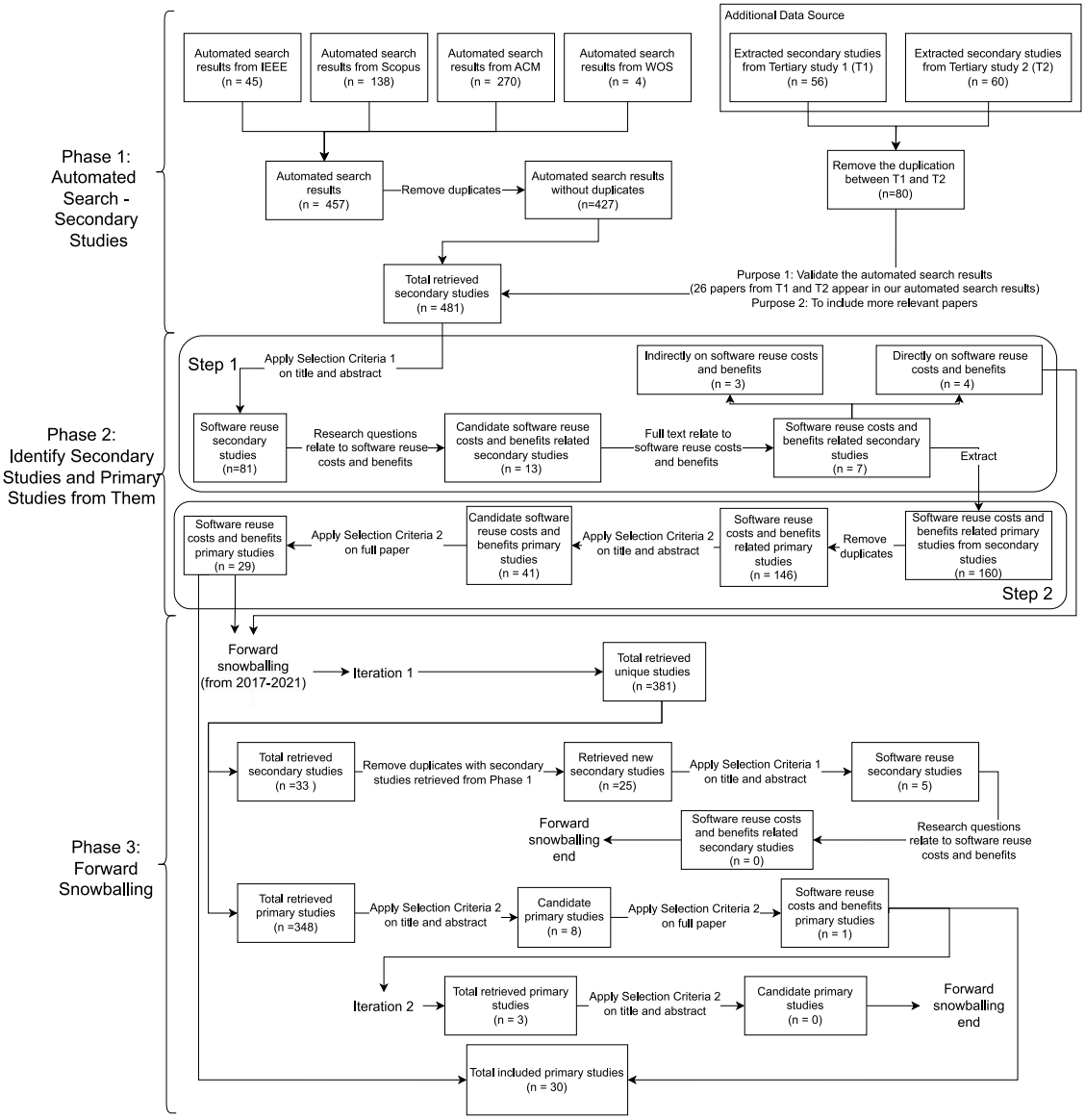

Fig. 1. Combined search strategy and primary study selection process.

snowballing to cover the missing years from the included secondary studies. The included secondary studies include the primary studies published until the end of 2016, resulting in missing years from 2017 to 2021. We used Google Scholar as our forward snowballing search engine because of its comprehensive coverage. The new snowballing iteration starts when we finalize the included primary studies in the previous iteration. The snowballing iteration ends when no new studies are found.

We had two iterations and retrieved 360 primary studies and 25 new secondary studies in Iteration 1 and three primary studies in Iteration 2. The first author read the full text and independently applied the selection criteria to the retrieved secondary and primary studies. We included one more primary study from forward snowballing. Eventually, we included 30 primary studies on software reuse costs and benefits.

3.3. Selection criteria

In our study, we developed two selection criteria to identify secondary and primary studies on software reuse costs and benefits. For the Secondary Study Selection Criteria, we included the secondary studies that directly investigated software reuse, followed a systematic and structured process (i.e., SLR and SMS), were written in English, were not duplicated, and were published until 2021.

For the Primary Study Selection Criteria, we included the primary studies meeting all the following criteria.

1. Studies that are published in peer-reviewed workshops, conferences, and journals. And
2. Studies where software reuse costs and benefits are directly investigated. And
3. Studies that report quantitative or qualitative results based on cases from industry. And
4. When several studies reported the same study, we included only the most recent (normally the Journal version) paper.

3.4. Quality assessment

We used the quality assessment checklist developed by Dybå and Dingsøyr [32] as our quality assessment base, which is also recommended by Kitchenham et al. [33] and is originally from the quality assessment criteria on the critical appraisal skills program (CASP) [34].

We modified some questions in the quality assessment checklist[2] to fit our research topic.

The criteria of whether it is a research paper and the aims of the research in the quality assessment checklist are used to screen out irrelevant papers. The remaining criteria of the checklist were assessed on a four-point scale: "Yes", "Partial", "No", and "Not applicable". The partial scale was further elaborated in four levels: close to no - 0.25, close to yes - 0.75, and between yes and no - 0.5. We had four iterations for quality assessment piloting until we were satisfied and agreed on the conceptual understanding in the checklist. The changes we made in each iteration are as follows.

Iteration 1: We randomly selected three primary studies to pilot the quality assessment checklist. However, we noted some differences and made three improvements as follows.

Improvement 1: Assess both the criteria and their sub-questions to enhance the traceability of the assessment decision.

Improvement 2: Update context criteria using five context facets — product, processes, people, organization, and practices/tools/techniques, based on Petersen and Wohlin's work [35]. We omitted the context facet — market, as it is not frequently reported in industrial software engineering research [35] and is less relevant to our study objectives.

Improvement 3: Add one sub-question about the appropriateness of the research method used in the primary studies.

Iteration 2: We updated the quality assessment checklist and piloted it on a randomly selected primary study, excluding the ones we selected in Iteration 1. However, the first author excluded the study since it was an experience report, while the second author did not. We discussed this misalignment and agreed to include experience reports since they include industrial cases and are focusing on software reuse costs and benefits. As such, we did not exclude any primary studies in Iteration 2. We also agreed to treat the experience report differently, including their quality assessment criteria — less restriction on methodology-related criteria.

Iteration 3: We updated the quality assessment checklist and piloted it on another randomly selected primary study, excluding the ones we selected in Iterations 1 and 2. Both the first and second authors agreed on the quality assessment results. The second author reviewed the checklist again and agreed that we had a shared understanding, and the first author could start independently applying it.

Iteration 4: When the first author completed the quality assessment, the third author verified the results by applying the quality assessment checklist to three randomly selected papers. There were minor differences in quality assessment results, and the third author gave feedback on assessing the study motivation, sample size, researchers' bias on the study, and contribution to the field. According to the feedback, the first author reapplied the quality assessment checklist to all included primary studies.

3.5. Data extraction

We designed the data extraction form[3] based on the general items (e.g., year and author) and items corresponding to each RQ (e.g., reuse type). Two authors piloted the data extraction on three randomly selected primary studies to see if we could extract all the needed data and check the extracted data consistency. After discussing the piloting results, the first author independently applied the data extraction form to the rest of the primary studies. The second and third authors verified the extraction results by applying the data extraction form to three randomly selected primary studies. We discussed the verification results in a joint meeting, finding no major disagreements. However, we noticed that we extracted the metrics but not their results, e.g., we extracted metric – defect density but missed extracting its results – the investigated case got 24% defect reduction after adopting software reuse practices.

We then included data about the reuse comparison scheme, measurement level, and measurement results to address the missing data item. The first author independently extracted the new data items. The second author reviewed the extraction results and found inconsistent granularity issues in reporting the measurement results. Some of the extracted measurement results are at a high level, such as the productivity increased after reuse, while others are at a finer level, such as the productivity increased 25% after reuse. The first and second authors discussed and agreed that a finer level of data extraction provides more information to the data analysis. The first author re-extracted the data to a finer level, which the second author reviewed.

3.6. Data analysis

We synthesized the extracted data to answer the RQs. For RQ1, we analyzed the data according to three categories: general information, context characteristics, and reuse-specific characteristics. For RQ2, we used thematic analysis [36] to code the software reuse costs and benefits. We also identified patterns by mapping the software reuse costs and benefits to different characteristics, such as reuse methods. We grouped and analyzed the metrics according to each software reuse cost and benefit in RQ3. For RQ4, we analyzed the quality assessment results based on different types of study — qualitative study, quantitative study, and experience report. We used narrative synthesis [37] to understand how validity threats are reported in the primary studies and analyzed them based on four categories: construct validity, conclusion validity, internal validity, and external validity. The first author conducted the preliminary data analysis, and the results were reviewed and discussed iteratively with the second author. The third author reviewed the final results, and we addressed the review comments in a joint meeting.

3.7. Validity threats

We followed the guidelines developed by Ampatzoglou et al. [38] on reporting the validity threats of secondary studies in software engineering.

- **Study selection validity** concerns the threats that occur in the study search and selection processes. We used automated search to identify secondary studies on software reuse costs and benefits and extracted the primary studies on software reuse costs and benefits from industrial cases as our start set for snowballing. To mitigate the threat of missing secondary studies, we used four databases and searched for all secondary studies on software reuse to make our search more inclusive. We also used two tertiary studies as our additional data source to include more papers. As for the threat of missing primary studies, we used forward snowballing on Google Scholar to cover the missing years by the selected secondary studies. Since the identified secondary studies have gone through a thorough search process — automated search, manual search, and snowballing, which covers the primary studies published before 2017, the chances of missing primary studies before 2017 are low. To avoid including the wrong papers, all three authors had a shared understanding of the selection criteria and piloted the selection criteria on randomly selected secondary/primary studies. The first author applied the selection criteria to all primary studies, and the second author reviewed all selection results.

[2] The final quality assessment checklist is available at https://zenodo.org/records/10649264.

[3] The detailed data extraction form is available at https://zenodo.org/records/10649264.

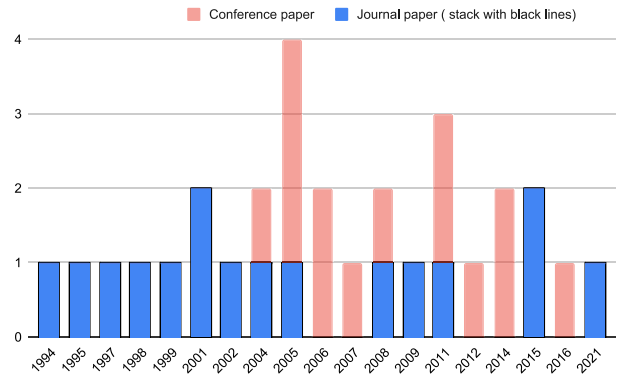

Fig. 2. Number of primary papers per year.

Table 2
Number of primary studies in different study types[a].

Study types	Primary studies
QN	17 (QN1 [1], QN3 [12], QN4 [2], QN5 [23], QN7 [41], QN9 [13], QN10 [4], QN11 [3], QN12 [15], QN13 [42], QN15 [21], QN21 [43], QN23 [44], QN25 [45], QN26 [46], QN28 [47], QN29 [48])
ER	10 (ER2 [7], ER6 [14], ER8 [10], ER16 [5], ER17 [9], ER18 [49], ER19 [8], ER22 [50], ER24 [6], ER27 [51])
QL	2 (QL14 [16], QL30 [52])
QNL	1 (QNL20 [53])

[a] The mapping between the paper ID and primary studies is available at https://zenodo.org/records/10649264

- **Data validity** involves threats related to the data extraction and analysis phase. The development of the data extraction form and the accuracy of the extracted data are our threats in the data extraction phase. To mitigate that, we involved two authors in developing the data extraction form according to the RQs. All authors have piloted the data extraction results on three randomly selected primary studies to have a shared understanding. The first and second authors customized the quality assessment checklist based on several piloting results. When the first and second authors had a consensus on the quality assessment checklist, the third author also piloted the quality assessment checklist on three randomly selected primary studies with the first author. In addition, we tried our best to provide an unbiased review. The conclusion reflects our knowledge of software reuse costs and benefits. The first author performed the main analysis, and the results were discussed with the other authors.
- **Research validity** includes threats that can be identified in all phases and concern the overall research design. We faced threats related to the study's replicability, the gap that the study addressed, the publication bias of the existing primary studies on software reuse costs and benefits, and the authors' familiarity with the research field. To ensure the study's replicability, we developed the study protocol according to Kitchenham et al. [33]. Based on our literature search on software reuse secondary studies, we identified a need to investigate the state of the art of software reuse costs and benefits. We found relatively fewer studies on software reuse costs, which may result in an unclear tradeoff between software reuse costs and benefits. All authors have worked with industrial software reuse, and we tried our best to provide an objective review when analyzing the studies.

4. Characteristics of the included primary studies (RQ1)

The results are based on 30 primary studies. This section describes the primary studies from three perspectives: (1) general information of the primary studies, (2) contextual characteristics, and (3) reuse-specific characteristics.

4.1. General information of the primary studies

The general information of the primary studies includes publication type, publication year, publication channels, and study type.

14 out of 30 primary studies were published at conferences, while the remaining 16 were published as journal papers. Table 3 shows the publication type and channels of the selected primary studies. Fig. 2 shows the number of primary studies over the years. We found that software reuse costs and benefits have been investigated in industrial cases the most between 2001 and 2015. There was a publication gap between 2017 and 2020. However, we did find two primary studies [39,40] specifically investigated software reuse costs and benefits and were published in 2019 and 2020 during our snowballing search process. We excluded them because the studies did not include industrial cases.

All primary studies have one or more industrial cases where software reuse was practiced. We categorized them into three study types: quantitative studies, experience reports, and qualitative studies. In total, we had 17 quantitative studies (QN), ten experience reports (ER), two qualitative studies (QL), and one both qualitative and quantitative (QNL) study (see Table 2).

4.2. Contextual characteristics

Contextual characteristics include the context information about the investigated industrial case (i.e., the type of development methodology and programming languages and the size of the company and the project). We summarized the contextual characteristics results in Table 4.

The results show that only five out of 30 primary studies reported the development methodology. Mohagheghi et al. (QN9 [13]), Mohagheghi and Conradi (QN13 [42]), Kolb et al. (QN23 [44]), one of the divisions in Lim (ER2 [7]) explicitly mentioned they followed incremental development. Gupta et al. (QN21 [43]) mentioned the waterfall process for one of the studied industrial cases.

Fifteen primary studies (50%) have described the company sizes, of which seven investigated large-sized companies, six investigated medium-sized companies, and three investigated small-sized companies. The results show that most studies investigated large- and medium-sized companies (more than 80%) instead of small-sized companies. More empirical studies are needed to investigate whether software reuse benefits small-sized companies the same as large- and medium-sized companies.

Twenty-four primary studies (80%) have mentioned their project sizes. The results show that software reuse costs and benefits were evaluated relatively uniformly across all ranges of project sizes — seven small-scale projects, nine medium-scale projects, and eight large-scale projects.

Seventeen primary studies (57%) reported the programming languages used in the investigated reuse cases, of which C (eight primary studies) and Java (four primary studies) were studied the most.

4.3. Reuse-specific characteristics

Reuse-specific characteristics refer to the information typically related to software reuse in industrial cases investigating costs and benefits, which includes reuse approach, reuse type, reuse purpose, reuse method, reuse comparison scheme, and measurement level. Table 5 presents the reuse-specific characteristics.

Table 3
Publication type and channels.

Publication type	Publication channels	Primary studies
Journal	IEEE Transactions on Software Engineering	1994 [QN1], 1998 [QN4], 1999 [QN5], 2004 [ER8]
	IEEE Software	1995 [ER2], 1995 [ER19]
	Journal of Systems and Software	1997 [QN3], 2001 [QN7]
	Software Process: Improvement and Practice	1998 [ER27]
	Decision Sciences	1999 [QN28]
	ACM Transactions on Software Engineering and Methodology	2008 [QN13]
	Empirical Software Engineering	2009 [QN21]
	Journal of Software Engineering and Applications	2011 [QL30]
	Requirements Engineering	2015 [QNL20]
	Total Quality Management and Business Excellence	2015 [QN25]
	Business & Information Systems Engineering	2021 [QN29]
Conference	IEEE International Conference on Software Engineering	2004 [QN9]
	IEEE International Conference on Software - Science, Technology and Engineering	2005 [ER6]
	IEEE International Conference on Software Maintenance	2005 [QN10]
	Springer International Conference on Software Product Line	2005 [QN11]
	IEEE International Software Product Line Conference	2006 [QN23] , 2007 [ER18], 2011 [ER16]
	IEEE International Conference on Management of Innovation and Technology	2012 [QN12]
	ACM-IEEE International Symposium on Empirical Software Engineering	2006 [QL14]
	Springer International Conference on Software Reuse	2008 [ER17]
	Springer International Conference on Computational Science and Its Applications	2014 [QN15]
	ACM International Software Product Line Conference	2011 [ER24], 2014 [ER22]
	IEEE/ACIS International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing	2016 [QN26]

Table 4
Contextual characteristics of the primary studies.

Contextual characteristics	Results: are presented as - Extracted characteristic (Number of primary studies reported) [References]
Development methodology[a]	- Incremental development (4) [ER2, QN9, QN13, QN23] - Waterfall development (1) [QN21]
Company size[b]	- Large-sized companies (7) [QN9, QN13, QL14, QN21, QN23, ER24, QN7] - Medium-sized companies (6) [QN1, ER6, QN7, QN10, ER18, QN28] - Small-sized companies (3) [QN12, ER17, QN25]
Project size[c]	- Small-scale projects (7) [QN4, ER6, QN7, ER8, QN11, QN12, QN15] - Medium-scale projects (9) [ER2, QN1, QN10, QN21, QN25, QN28, QL14, ER17, ER22] - Large-scale projects (8) [QN3, QN5, QN9, QN13, QN26, ER19, QNL20, ER24]
Programming languages	- C (8) [ER2, QN7, QN9, QN12, QN13, QN23, ER24, QN25] - Java (4) [QN4, QN13, QL14, QN21] - RPG (2) [QN1, QN11] - C++ (2) [QN7, ER19] - Erlang (2) [QN9, QN13] - Ada (1) [QN3] - Pascal (1) [ER2] - Fortran (1) [QN5] - Perl (1) [QN9] - COBOL (1) [QN10]

[a] The development methodology stands for what software development process is followed in the case companies, such as agile development, continuous integration, and incremental development.

[b] We divided the reported companies into three scales in accordance with the European Commission's classification of micro, small, and medium-sized companies [54]: large-sized companies have more than 250 employees; medium-sized companies have more than 50 but less than 250 employees; and small-sized companies have less than 50 employees.

[c] We followed the same project size category as Mohagheghi and Conradi [17]. Small-scale objects: a few reused software assets or small products; medium-scale objects: larger than the small-scaled ones, but the objects are still few, or the software size is less or around 100 KLOC; large-scale objects: the objects are quite a few, or the software size is more than 100 KLOC.

The reuse approach refers to the technical methods used by industrial cases to implement software reuse. We identified eight reuse approaches from the primary studies: software product line (SPL), reuse from repository, compositional reuse, component reuse, framework reuse, template reuse, engine reuse, and a mixed reuse approach between architecture reuse and context-dependent module reuse. The results show that SPL is the most used reuse approach among the investigated industrial cases, accounting for 47% of all reported reuse approaches in the primary studies. The second most used reuse approach is reuse from repository, and the rest of the reuse approaches are reported once or twice.

There are two types of code asset reuse: verbatim reuse (reuse as-is or black-box reuse) and modified reuse (white-box reuse). The results show that both reuse types are widely used in the primary studies. Nineteen primary studies reported verbatim reuse, and 14 reported modified reuse. In addition, industrial cases in ten primary studies adopted both verbatim and modified reuse.

The reusable assets can be used for different purposes, i.e., infrastructure reuse, domain reuse, and architecture reuse [17]. The results show that 75% reported reusable assets are designed for a specific domain, and fewer are about the infrastructure (21%) and architecture (4%). This result is expected since the most reported reuse approach is SPL, which comprises a product family within a specific application domain.

The reuse method refers to whether the company follows a systematic way to implement software reuse, including systematic reuse, ad

Table 5
Reuse-specific characteristics of the primary studies.

Reuse-specific characteristics	Results: are presented as - Extracted characteristic (Number of primary studies reported) [References]
Reuse approach	- SPL[a](14) [QN3, ER8, QN9, QN11, QN13, QN15, ER16, ER17, ER18, ER19, ER22, QN23, ER24, QN29] - Reuse from repository (6) [ER6, QN10, QN12, QNL20, QN25, QN28] - Compositional[b](2) [QN1, QN7] - Component reuse[c] (1) [QL14] - Framework reuse[d] (2) [QN4, QN21] - Template reuse[e](2) [QN10, QN26] - Engine reuse[f](1) [QN15] - Architecture reuse and context-dependent module reuse (2) [QN5, QL30]
Reuse type	- Verbatim (black-box) reuse (19) [QN1, QN15, QN3, QN4, QN5, ER6, QN7, ER8, QN9, QN10, QN12, QN13, QL14, ER19, QNL20, QN21, QN25, ER27, QL30] - Modified (white-box) reuse (14) [ER2, QN3, QN4, QN5, ER8, QN10, QN11, QL14, ER19, QN25, QN26, ER27, QN29, QL30]
Reuse purpose	- Domain reuse[g](18) [QN1, QN3, QN4, QN5, QN9, QN10, QN11, QN12, QN13, QN15, ER17, ER18, ER19, QN21, QN23, ER24, QN25, QN29] - Infrastructure reuse[h](5) [QN1, QN13, ER19, QNL20, QN21] - Architecture reuse[i](1) [QN23]
Reuse method	- Systematic reuse[j](20) [QN1, ER2, QN3, QN4, QN5, ER6, ER8, QN9, QN10, QN13, ER17, ER18, ER19, QNL20, QN21, ER22, QN23, ER24, ER27, QN28] - Ad hoc reuse[k](4) [QN1, ER17, ER19, QN7] - Controlled reuse[l](3) [ER8, QN12, QN25]
Reuse comparison scheme	- Comparing the software reuse impact between the original project developed from scratch and the projects that reuse the original project (8) [ER19, QNL20, QN21, ER22, QN23, QN9, ER27, QN29] - Comparing the software reuse impact between the assets that follow and do not follow a reuse approach (6) [QN1, QN4, QN10, QN11, ER16, ER24] - In the same projects, comparing the software reuse impact between the reused assets and new assets (3) [ER2, QN9, QN13] - Comparing the software reuse impact between the assets based on reuse method - systematic reuse, ad hoc reuse, controlled reuse, and pure development (3) [ER6, ER8, ER17] - Comparing the software reuse impact between different types of modified reused assets - newly developed assets, slightly modified assets, extensively modified assets, and reuse verbatim assets (2) [QN3, QN5] - Comparing the software reuse impact between the assets with different reuse rates (5) [QN7, QN12, QN15, QN25, QN28]
Measurement level	- Project/system/asset/product level (22) [QN1, ER2, QN4, ER6, ER8, QN10, QN12, QN15, ER16, ER17, ER18, ER19, QNL20, QN21, ER22, QN23, ER24, QN25, QN26, ER27, QN28, QN29] - Module/component level (7) [QN3, QN5, QN7, QN9, QN11, QN13, QN15]

[a] Software product line (SPL) uses domain engineering to develop domain-specific reusable assets, and all products derived from the SPL reuse the same architecture. Selecting and reusing reusable assets from repositories is another reuse approach in the investigated cases of the primary studies.
[b] Compositional reuse stands for reusing both assets that are specific to a single product and generic to various projects.
[c] Component reuse represents reusing components from existing systems.
[d] The framework contains pre-written design patterns and components of common functionalities that provide a foundation for building the system.
[e] The reused template has a basic structure of the assets, which includes a pre-determined part that is independent from the system and a blank part that engineers use to develop explicitly for the system.
[f] Engine reuse, or transformation-based reuse, refers to reuse engines that can produce outputs by transforming appropriate inputs.
[g] Infrastructure reuse refers to reusing common assets across different domains, such as the login function.
[h] Domain-specific reuse stands for reusing common assets with a specific domain.
[i] Architecture reuse refers to reusing the structures to build a system. 19 primary studies reported the purpose of developing or using reusable assets.
[j] Systematic reuse aims to develop new systems using existing software assets in an organized and planned way.
[k] Ad hoc reuse, on the other hand, follows no formalized process or guidelines when implementing software reuse.
[l] Controlled reuse involves a more structured approach than ad hoc reuse but is less formal than systematic reuse.

hoc reuse, and controlled reuse. Twenty-three primary studies mentioned reuse methods, of which 18 mentioned a single reuse method, and four reported multiple reuse methods. In these four primary studies, three primary studies investigated both ad hoc reuse and systematic reuse on different projects (QN1 [1], ER17 [9], ER19 [8]), and one primary study investigated both controlled reuse and systematic reuse on different companies (ER8 [10]). Twenty primary studies (74%) focused on systematic reuse, while only four and three studies reported ad hoc reuse and controlled reuse.

In our sample, we found authors have used six different reuse comparison schemes to evaluate to what extent software reuse results in benefits or costs in the industrial cases. 14 primary studies compared software reuse costs and benefits between projects, of which eight compared the original project (which developed from scratch) and the projects reused the original ones, and six compared the projects that followed a reuse approach and those that did not. Three primary studies (ER2 [7], QN9 [13], QN13 [42]) compared the costs and benefits of reused and new assets in the same project. Three primary studies (ER6 [14], ER8 [10], ER17 [9]) compared the costs and benefits of the same projects using different reuse methods. Two primary studies (QN3 [12], QN5 [23]) compared the costs and benefits of the assets with different reuse modification levels. Five primary studies (QN7 [41], QN12 [15], QN15 [21], QN25 [45], QN28 [47]) used the reuse rate as the basis of comparison to investigate the costs and benefits.

The costs and benefits of software reuse can be measured at the high (project) and low (component) levels. The results show that the industrial cases have measured software reuse costs and benefits more on the high level than the low level - 22 compared to 7.

RQ1 Key findings: We included 30 primary studies and extracted four context characteristics and six reuse-specific characteristics from our study sample. The key findings about the extracted characteristics are as follows.

- Most primary studies on software reuse costs and benefits were conducted within medium- and large-sized companies (13 primary studies), while only a few investigated small-sized companies (three primary studies).
- The primary studies have investigated software reuse costs and benefits almost equally in all types of project sizes: small-scale projects (seven primary studies), medium-scale projects (nine primary studies), and large-scale projects (eight primary studies).

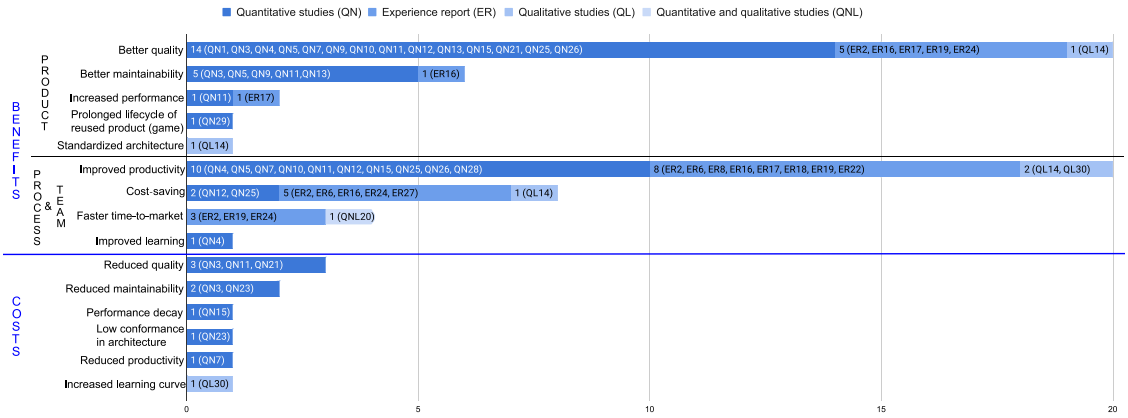

Fig. 3. Software reuse costs and benefits identified from the primary studies.

- Programming Languages C (eight primary studies) and Java (four primary studies) were studied the most in our sample.
- Only five primary studies in our sample reported the development methodology. Therefore, it is unclear in which context the other included studies practiced software reuse.
- SPL (14 primary studies) and reuse from a repository (six primary studies) are the most investigated reuse approaches in our sample.
- Most primary studies used verbatim reuse (nine primary studies) or both verbatim and modified reuse (ten primary studies). Only a few used modified reuse (four primary studies).
- Systematic reuse (20 primary studies) was investigated the most in our sample, and only a few primary studies investigated ad hoc (four primary studies) and controlled reuse (three primary studies).
- Our sample investigated more domain reuse (18 primary studies) than infrastructure (five primary studies) and architecture reuse (one primary study). This finding aligns with the finding in reuse approaches that the most investigated reuse approach is SPL.
- The most investigated comparison schemes in our sample are the comparison of original projects versus projects that reused them (eight primary studies), assets following a reuse approach versus those that did not (six primary studies), and assets with varying reuse rates (five primary studies).
- Most primary studies measured reuse impacts on high levels - e.g., project level (22 primary studies) rather than low levels - e.g., module level (seven primary studies).

5. Software reuse costs and benefits and their relation with reuse practices (RQ2)

We identified nine software reuse benefits and six software reuse costs from primary studies (see Fig. 3). Software reuse benefits (29 primary studies) were investigated more than costs (seven primary studies). The most investigated benefits in our sample are better quality (20 out of 30 primary studies) and improved productivity (20 out of 30 primary studies). More than three studies reported software reuse benefits in cost-saving, better maintainability, and faster time-to-market. The rest of the benefits are mentioned once or twice. As for the reported software reuse costs, only reduced quality and reduced maintainability were reported three or two times. The rest of the costs are reported only once. The contradicting findings, e.g., better quality and reduced quality, are further explained in metric level in RQ3 (see Section 6.2).

We divided the benefits into two types of improvements: product-related benefits and process and team related benefits. Product-related benefits include better quality, better maintainability, increased performance, and prolonged lifecycle of reused product (game)[4] and standardized architecture. Process and team related benefits include improved productivity, cost-saving, faster time-to-market, and improved learning.

We also found that quantitative studies and experience reports reported the most benefits — better quality, improved productivity, cost-saving, better maintainability, and faster time-to-market. All reported software reuse costs were reported in quantitative studies, except for the increased learning curve. The results show that the stand-out software reuse benefits are better quality and improved productivity.

We mapped some characteristics — reuse approach, reuse type, project size, reuse purpose, reuse method, and measurement level, with the identified software reuse impacts (costs and benefits) to see which characteristics are more likely to benefit companies. The relation between the identified software reuse impacts and reuse approach, reuse type, and reuse method stands out among the other characteristics.[5] Fig. 4 depicts the relation between the observed software reuse benefits and costs, and investigated software reuse approaches. The primary studies investigated SPL and reuse from repository the most (see Table 5). The mapping results between reuse approaches and software reuse benefits indicate that SPL covers more (seven out of nine) benefits than the other reuse approaches. Better quality has been observed in all identified reuse approaches. Improved productivity has been observed in all reuse approaches except engine reuse. Considering the number of primary studies and the coverage of benefits investigated in reuse approaches, SPL and reuse from repository are more likely to benefit companies. Within the limited primary studies on software reuse costs, the results showed that SPL is associated with some costs.

Fig. 5 shows the mapping between the observed software reuse benefits and costs, and reuse types and methods. For reuse types, the results show that in our sample, most benefits were observed when both verbatim and modified reuse were applied. In the cases of using a single reuse type, more primary studies investigated verbatim reuse (nine primary studies) than modified reuse (four primary studies). By looking at those primary studies that only adopt modified or verbatim reuse, the results show that verbatim reuse may result in more benefits to companies, considering the number of primary studies and the coverage of benefits investigated in verbatim/modified reuse. These findings were also observed in primary studies that explicitly investigated the costs and benefits among different reuse types: Thomas et al. (QN3 [12]), and

[4] Game B reuses components from Game A. Such reuse helped prolong the lifecycle of Game A.

[5] The relations between the identified software reuse impacts (costs and benefits) and project size, reuse purpose, and measurement level are available at https://zenodo.org/records/10649264.

Fig. 4. Software reuse costs, benefits and reuse approaches.

Selby (QN5 [23]) found that verbatim reuse has a more positive impact than modified reuse on quality, maintainability, and productivity.

As for the reuse methods, most primary studies have adopted systematic reuse. All reported reuse methods have benefits in better quality and improved productivity. The results show that systematic reuse has more reuse benefits compared to controlled and ad hoc reuse. Similar findings were identified in three primary studies that explicitly investigated the costs and benefits among different reuse methods: Morad and Kuflik (ER6 [14]) and Tomer et al. (ER8 [10]) found systematic reuse had better productivity in all investigated code assets except one. Beyer et al. (ER17 [9]) found systematic reuse resulted in better quality, productivity, and performance when compared to ad hoc reuse.

RQ2 Key findings: Software reuse benefits (29 primary studies) have been investigated more than the costs (seven primary studies) in our sample. The following are the key findings about the identified software reuse costs and benefits.

- Better quality (20 primary studies) and improved productivity (20 primary studies) were the most frequently investigated software reuse benefits in our sample.
- Reduced quality (three primary studies) was the most frequently investigated software reuse costs in our sample. The reasons for software reuse having both positive and negative quality impacts are further explained in RQ3 (see Section 6.2).
- Our results showed that SPL (seven benefits and 14 primary studies) is more likely to benefit companies.
- Our results showed that systematic reuse (seven benefits and 20 primary studies) is more likely to benefit companies.
- Our results showed that verbatim reuse (five benefits and nine primary studies) is more likely to benefit companies.
- Better quality and improved productivity are the software reuse benefits generally observed in all identified reuse approaches.

6. Metrics for measuring software reuse costs and benefits (RQ3)

This section presents the metrics that the companies used to measure software reuse costs and benefits. We cluster the metrics according to the identified software reuse costs and benefits from RQ2. We present the metrics, their definitions, and results in an aggregated form.[6]

6.1. Metrics for software reuse benefits

This sub-section presents the metrics used to measure the identified software reuse benefits.

6.1.1. Metrics for better quality

Twenty primary studies identified better quality as a benefit of software reuse. Moreover, 19 of these primary studies used 12 unique metrics to measure quality benefits while practicing software reuse (see details in Table 6). The most used metrics relate to code problems, such as defects and bugs. Some metrics relate to code structure, developers' subjective rating on debugging and maintaining software, and customer complaints. Defect density is the most frequently used metric. We noticed that primary studies used different terms when describing defect density, such as fault density and error density. We refer to them as defect density in the rest of the paragraph. Some primary studies measured defect density and found software reuse helped reduce the defects (ER2 [7], QN9 [13], QN13 [42], QN26 [46]) and some found higher reuse rate and verbatim reuse resulted in fewer defects (QN3 [12], QN5 [23], QN7 [41], QN12 [15], QN25 [45]). In addition, Gupta et al. (QN21 [43]) found that software reuse helps decrease the defect density of the reused software. Three primary studies counted the number of defects/bugs to investigate better quality in practicing software reuse. Deniz and Bilgen (QN15 [21]) counted the number of defects of the reused components in different products and found that the quality of reused components improved with more reuse. On the other hand, Deniz and Bilgen (QN15 [21]), Quilty and Cinneide (ER16 [5]), and Otsuka et al. (ER24 [6]) used the defect counts to evaluate the quality between reused and non-reused projects or products. However, they did not normalize the number of defects with the project size, which makes

[6] The detailed metrics definitions and results of each primary study are available at https://zenodo.org/records/10649264.

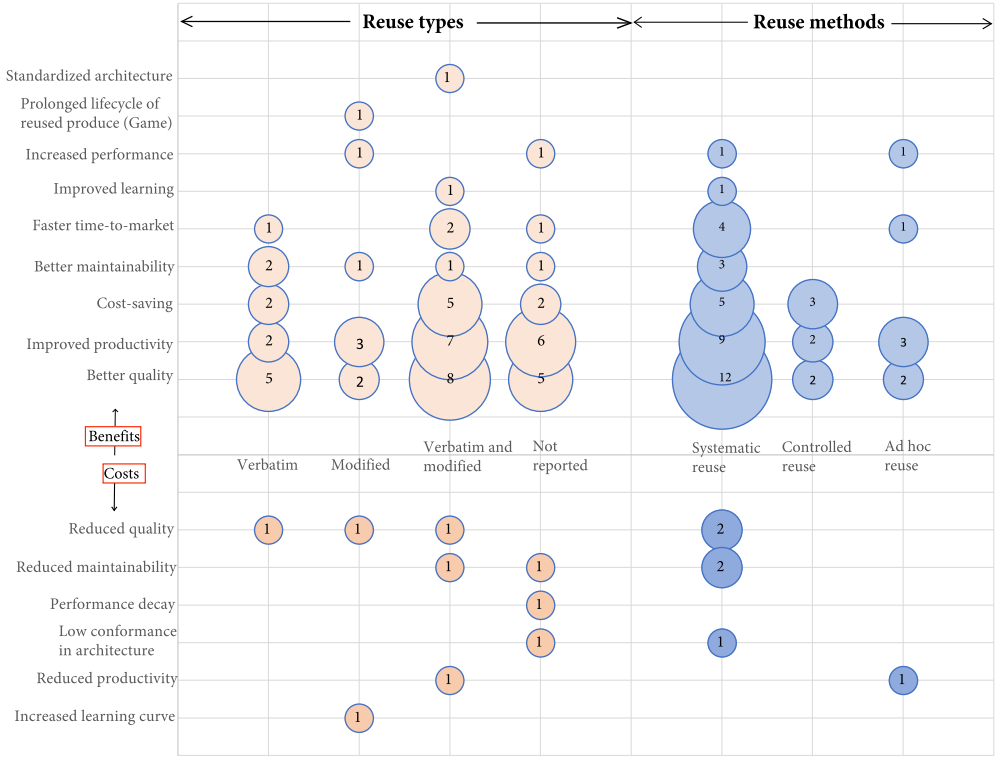

Fig. 5. Software reuse costs, benefits, reuse types, and reuse methods.

the result less comparable. Multiple primary studies used complexity and coupling metrics. The results showed that software reuse helps in reducing the code complexity (QN10 [4], QN15 [21], ER17 [9]) and improving the code coupling (QN11 [3], QN15 [21]). The rest of the quality metrics are only used by a single primary study (see details in Table 6).

6.1.2. Metrics for improved development productivity

Twenty primary studies identified improved productivity as a benefit of software reuse. Moreover, 18 of these primary studies used eight unique metrics to measure productivity gains while practicing software reuse (see details in Table 7). Most productivity metrics relate to development effort, while some of them relate to effort in requirements (QN15 [21]), function points (QN4 [2]), design (QN5 [23]), or testing (ER17 [9]). We found four kinds of productivity metrics related to development effort: the size of the assets per development effort (size per effort), the development effort per the size of the assets (effort per size), the total development effort, and the total size of the developed assets. The most used productivity metrics are related to size per effort. Depending on whether to include the size of reused assets in the total size of the assets, productivity (size per effort) can be further divided into apparent/gross productivity (total size of the written code and reused code per development effort) and net/actual productivity (size of the written code per development effort). Ten primary studies measured apparent/gross productivity, and three measured net/actual productivity. By using apparent/gross productivity, the primary studies found (1) software reuse increases productivity (ER2 [7], QN10 [4], QN15 [21], ER19 [8], ER22 [50], QN26 [46]), and (2) the higher reuse rate, the more increased in productivity (QN7 [41], QN12 [15], QN25 [45], QN28 [47]). Morisio et al. (QN4 [2]) and Deniz and Bilgen (QN15 [21]) used actual productivity and found that software reuse helps improve productivity. Baldassarre et al. (QN10 [4]) investigated actual productivity in different periods of reuse approaches and found systematic reuse helps increase productivity in the early phase when developing for reuse. When both the non-reuse and reuse processes mature, their actual productivity tends to range within a defined interval. Three primary studies used metrics of effort per size to measure productivity. The results showed that software reuse helps increase productivity (ER18 [49]) and the more reuse, the better productivity (QN5 [23], QN7 [41]). In addition to the normalized metrics, Selby (QN5 [23]) also used non-normalized metrics (development effort and code size) to measure productivity and found that verbatim reuse has the least development effort and smallest code size. Morad and Kuflik (ER6 [14]) and Tomer et al. (ER8 [10]) measured the development effort of using different reuse methods for the same assets and found systematic reuse required the least development effort. Three primary studies (QN11 [3], ER16 [5], ER17 [9]) investigated productivity only using a non-normalized metric — development effort and found software reuse helps reduce development effort.

6.1.3. Metrics for better maintainability

Six primary studies identified better maintainability as a benefit of software reuse. They used six unique metrics to measure better maintainability when practicing software reuse (see details in Table 8). Maintainability is not limited to code changes but their required effort. The six metrics can be categorized into maintenance effort (QN3 [12], QN5 [23], ER16 [5]), modification rate (QN5 [23], QN9 [13], QN13 [42]) and size of the reduced code (QN11 [3]). Only metrics — fault correction effort and code modification rate, were used by more than one primary study. The results of fault correction effort showed projects following software reuse have less effort spent on bug fixing (ER16 [5]) and verbatim reuse has the lowest effort in fault correction (QN5 [23]). Mohagheghi et al. (QN9 [13]), Mohagheghi and Conradi (QN13 [42]), and Selby et al. (QN5 [23]) found reused components are less changed. Selby et al. (QN5 [23]) found that verbatim reuse has the lowest fault isolation effort, change correction effort, change density, and average number of changes per module. Zhang and Jarzabek (QN11 [3]) found that reuse helps reduce the number of codes compared to the original code size. Thomas et al. (QN3 [12]) found that verbatim reused and slightly modified reused modules had less rework effort than newly created ones.

Table 6
Better quality metrics, definitions and results.

Metrics (number of primary studies used)	Definition from primary studies	Results
Defect/Fault/Error density: the number of defects/faults/errors divided by the software size (10)	- ER2 (1994), QN7 (2001), QN9 (2004), QN12 (2012), QN21 (2009), QN25 (2015): the number of defects/errors/ trouble reports per (thousand) non-commented source line of code. - QN3 (19997), QN5 (2005), QN13 (2008), QN26 (2016): Errors/faults/trouble reports/defects (thousand) source line of code.	- ER2, QN3, QN9, QN13, QN26 reported that the defect densities of software that were developed with reuse approaches have decreased. - QN21 reported that the reusable software had a defect reduction after it had been reused. - QN5 reported that verbatim reuse had the fewest defect density. - QN7 reported that the more external reuse, the fewer defect densities. - QN12, QN25 reported the higher reuse rate, the fewer defect densities.
Number of defects/bugs (4)	- QN15 (2014), ER16 (2011), QN24 (2011): the number of defects/bugs. - ER16 (2011): the number of bugs at different bug severity levels.	- QN15 reported a decrease in defect counts in reusable components after they have been reused in several other products. - QN16 and QN24 reported a decrease in defect counts in software that was developed with reuse approaches. - ER16 reported software that was developed with reuse approaches had less number of bugs at all bug severity level.
Complexity (3)	- QN10 (2005): Mean Cyclomatic Complexity of the system. - QN15 (2014): McCabe Cyclomatic Complexity (McCabeCC), Percent Branch Statements (% Branches), Weighted Methods for Class (WMC), Nested Block Depth (NBD). - ER17 (2008): McCabe maximum nesting of control structures, McCabe Cyclomatic Complexity (McCabeCC).	- QN10, QN15, and ER17 reported that software developed with reuse approaches had lower code complexity.
Coupling (2)	- QN11 (2005), QN15 (2014): Coupling Between Objects (CBO)	- QN11 and QN15 report that software developed with reuse approaches had lower code coupling.
Customer complaint density (1)	- QN1 (2001): The number of customer complaints per line of code.	- QN1 reported that software reuse helped decrease the customer complaint density.
Rate of error slippage	- QN3 (1997): Percentage of errors that escape unit test.	- QN3 reported that slightly modified reused components have significantly lower error slippage rates.
Index of quality of programming (1)	- QN4 (2002): Development effort (excluding rework effort) per total development effort (development effort + rework effort).	- QN4 reported that development with reuse was less error-prone than traditional development.
Average faults per module (1)	- QN5 (2005): Average (and standard deviations) for total faults in different types of reuse modules.	- QN5 reported that verbatim reused modules had fewer faults compared to modules that were newly developed, majorly revised, or slightly revised.
Number of Delta (1)	- QN7 (2001): Delta represents a change to the software, such as an enhancement or a repair, which are sometimes used to estimate error rates.	- QN7 reported that the higher the External Reuse Frequency or External Reuse Level, the lower the Delta.
Quality rating (1)	- QN7 (2001): Quality rating is a subjective ordinal measure from developers' view regarding debugging and maintaining software. The scale ranges from 1-worst to 10-best.	- QN7 reported that the higher the External Reuse Frequency or External Reuse Level, the higher the rating in quality.
Number of field defects reported by customers (1)	- ER17 (2008): The aggregated number of field defects reported by customers.	- ER17 reported that customers reported fewer defects when following systematic reuse compared to ad hoc reuse.
Fault rate (1)	- ER19 (1995): Number of major errors to total recorded faults at the time of delivery.	- ER19 reported software reuse helps improve the fault rate of systems at the time of delivery.

6.1.4. Metrics for cost-saving

Eight primary studies identified cost-saving as a benefit of software reuse. Moreover, seven of these primary studies used six unique metrics to measure cost-saving when practicing software reuse (see details in Table 9). Lim (ER2 [7]), Ha et al. (QN12 [15]), Sun et al. (QN25 [45]), and Powell and Brown (ER27 [51]) monetized the cost-saving, and they found reuse helps save cost. Morad and Kuflik (ER6 [14]), Quilty and Cinneide (ER16 [5]), and Otsuka et al. (ER24 [6]) found that the costs of implementing reusable assets will be paid off after several times of implementations.

However, compared to other extracted quality or productivity benefits related metrics, these monetized cost-saving metrics lack a clear definition. For example, Ha et al. (QN12 [15]) monetized the total software reuse costs using software reuse development costs and management and support costs. The software reuse development costs were calculated by multiplying the overall development effort by the salary rate of the developers. In contrast, the management and support related costs were not clearly defined. In another case, Quilty and Cinneide (ER16 [5]) looked for the breakeven point when the reuse benefits have paid off the costs of developing reusable assets. However, how the costs are calculated is not clear.

6.1.5. Metrics for faster time-to-market

Four primary studies identified faster time-to-market as a benefit of software reuse. Moreover, these four primary studies used a unique metric to measure faster time-to-market when practicing software reuse (see details in Table 10). All four primary studies tracked how long it took to develop an asset from beginning to end. Furthermore, the results show that software reuse helps faster time-to-delivery (ER2 [7], ER19 [8], QNL20 [53], ER24 [6]).

6.1.6. Metrics for other benefits

Table 11 presents four unique metrics used to verify the software reuse benefits in increased performance, improved learning, and prolonged lifecycle of reused product (game). Zhang and Jarzabek (QN11 [3]) and Beyer at al. (ER17 [9]) investigated the performance using metrics — running time and memory usage. They found software reuse can help faster the game processing time (QN11 [3]) and reduce memory usage (QN11 [3], ER17 [9]). Morisio et al. (QN4 [2]) investigated the developers' learning in development with and without reuse conditions, and they found that software reuse increases the developers' learning. Mihale-Wilson (QN29 [48]) used grid analysis and found that reuse from previous games can help prolong the lifespan of previous games.

Table 7
Better productivity metrics, definitions and results.

Metrics (number of primary studies used)	Definition from primary studies	Results
Apparent/Gross productivity: total size of project per total development effort (10)	- ER2 (1994), QN7 (2001), QN12 (2012), QN15 (2014), QN25 (2014): (thousand) non-comment line of code per engineering month/ person-day/person-hour. - QN10 (2005), QN26 (2016): Line of code per person-hour/person-day. - QN15 (2014): number of requirements per person-hour. - ER19 (1995): average line of code per person-day. - QN26 (2016): total number of Web pages per person-day. - QN28 (1999): The productivity is calculated in a combination of the productivity of new code, the productivity of reused code, the quality of the reuse decision, and the relative value of reuse to the company. - ER22 (2014): number of development models per year.	- QN7 reported that one case showed a positive relationship between productivity and External Reuse Level, and another case showed a marginally positive relationship between productivity and External Reuse Frequency. - ER2, QN10, QN15, ER19, QN26, QN28, and ER22 reported that productivity increased after applying software reuse. - QN12 and QN25 reported a positive relationship between productivity and reuse rate.
(Average) Development effort (6)	- QN5 (2005), ER6 (2005), ER8 (2004), QN11 (2005), ER17 (2008), ER16 (2011): Total development time is (tenths of) person hours/man-day/person-month.	- QN5 reported that newly developed and majorly revised modules had the most development effort, slightly revised modules had the second most, and those completely reused had the least. - ER6 and ER8 reported that systematic reuse had the least development effort compared to development from scratch and other reuse methods. - ER6, QN11, ER17, and ER16 reported that software reuse helped reduce the development effort.
Development effort per size of project (3)	- QN5 (2005): tenths of hours per line of code. - QN7 (2001): person-day per number of modules. - ER18 (2007): time (day or month) per product or effort (person-day) per product.	- QN5 reported that newly developed modules had the most effort per source line, majorly revised and slightly revised modules had the second most, and completely reused modules had the least. - QN7 reported that one case showed a negative relationship between effort/module and External Reuse Frequency and a marginally negative relationship between effort/module and External Reuse Level. Another case showed a negative relationship between External Reuse Frequency and effort/module. - QN18 reported that software reuse helped reduce the time and effort spent on each new product.
Net/Actual productivity (3)	- QN4 (2002): the net size of the developed application (object-oriented function points)/hours - QN10 (2005): written line of code per person-hour - QN15 (2014): number of new requirements per man-hour.	- QN4 reported that applications developed with reuse provide higher net productivity than the ones developed without reuse. - QN10 reported that reuse-oriented development helped increase the actual productivity in the early phase when developing for reuse. - QN15 reported that productivity improved when developers were familiar with the development of the product line.
Size of the code (1)	- QN5 (2005): line of code	- QN5 reported that verbatim reuse has the smallest size of code compared to reusing majorly revised modules, newly developed modules, and slightly revised modules.
Actual/Estimated development effort (1)	- ER22 (2014): Actual development effort: total man-hours (month or year) spent in developing the core assets and the number of models with software reuse. Estimated development effort: estimated total man-hours (month or year) spent in developing the core assets and the number of models without software reuse.	- ER22 reported that the actual development effort with software reuse was less than the estimated development effort without software reuse.
Percentage of development effort spent on design (1)	- QN5 (2005): The effort spent during module design activities divided by total module development effort.	- QN5 reported that newly developed modules had a higher percentage of effort spent in design than verbatim reused, slightly revised, and majorly revised modules.
Testing effort (1)	- ER17 (2008): total testing time (person-month).	- ER17 reported that software reuse helped save testing time.

6.2. Metrics for software reuse costs

We identified seven primary studies that reported five software reuse costs. Table 12 presents all the costs and their corresponding metrics. Reduced quality was measured according to the defect density per defect type/ defect severity/ impact attributes (QN21 [43]), complexity (QN11 [3]), error source, and error type (QN3 [12]). Both Thomas et al. (QN3 [12]) and Gupta et al. (QN21 [43]) investigated the characteristics of defect types in different reusable assets. In addition, Zhang and Jarzabek (QN11 [3]) found that complexity increased slightly due to reuse. Maintainability drops due to the higher percentage of errors that require more than one day to fix (QN3 [12]), code complexity, and structuredness (QN23 [44]). Deniz and Bilgen (QN15 [21]) used the number of cycles, time delay, and CPU usage to measure performance and found that software reuse can result in performance decay. Frakes and Succi (QN7 [41]) measured the productivity in two cases, using two metrics — size per effort and effort per size. The results showed a negative relationship between software reuse and productivity. Kolb et al. (QN23 [44]) investigated the architecture conformance between the products and the product line, and they found that developing reusable assets violates the defined product line architecture. In addition, the architecture of the products derived from the product line was not fully conformed with the product line architecture.

When looking at the software reuse costs, we found that some impact attributes were also identified as benefits. The different results were mainly because of the different metrics used to measure the impact of software reuse. According to the primary studies, software reuse helps improve the overall product quality in terms of defect density (ER2 [7], QN7 [41], QN9 [13], QN12 [15], QN21 [43], QN25 [45], QN3 [12], QN5 [23], QN13 [42], QN26 [46]) but may result in more severe defects (QN21 [43]). Baldassarre et al. (QN10 [4]), Deniz and Bilgen (QN15 [21]), and Beyer et al. (ER17 [9]) found that software reuse improves the complexity of the product quality by measuring the number of linearly independent paths within the code assets, while Zhang and Jarzabek (QN11 [3]) found that the complexity was increased a bit after using a reuse technique by measuring the number

Table 8
Better maintainability metrics, definitions, and results.

Metrics (number of primary studies used)	Definition from primary studies	Results
Rework effort (1)	- QN3 (1997): Dividing rework effort by the number of source statements.	- QN3 reported that verbatim reused modules required less rework effort compared to other reuse types.
Fault correction effort (2)	- QN5 (2005): Line of code in tenths of hours in fault correction development effort. - ER16 (2011): Hours spent on bug fixing by year.	- QN5 reported that verbatim reused modules had the lowest fault correction effort. - ER16 reported that the overall required maintenance effort was reduced after practicing software reuse.
Code modification rate (MOD)/Change density (3)	- QN9 (2004), QN13 (2008), QN5 (2005): Size of modified (or new) code divided by the total size of the code. - QN5 (2005): (Average) Number of changes per module.	- QN9, QN13, QN5 reported that reused components were less modified (or changed) than non-reused ones. - QN5 reported that verbatim reused modules had the least changes per source line.
Fault isolation effort (1)	- QN5 (2005): Line of code in tenths of hours in fault isolation development effort.	- QN5 reported that verbatim reused modules had the lowest fault isolation effort.
Change correction effort (1)	- QN5 (2005): Line of code in tenths of hours in change correction development effort.	- QN5 reported that verbatim reused modules had the lowest change correction effort.
Number of code reduced (1)	- QN11 (2005): The size of the reduced code compared to the original size of the code.	- QN11 reported that the code size was reduced compared to the original code size after software reuse.

Table 9
Cost-saving metrics, definitions, and results.

Metrics (number of primary studies used)	Definition from primary studies	Results
Net-present-value (1)	- ER2 (1994): Net present value takes the estimated value of reuse benefits and subtracts from its associated costs, taking into account the time value of money.	- ER2 reported that both cases saved a lot after implementing software reuse.
Savings of systematic reuse over new development (1)	- ER6 (2005): The percentage of savings between systematic reuse over new development.	- ER6 reported that both projects showed that software reuse saves more than new development.
Costs (2)	- QN12 (2012), and QN25 (2015): The total cost of developing software using software reuse is the total development effort (developing reusable assets and software products) multiplied by the salary rate of the developer plus costs in management and support.	- QN12 and QN25 reported that the cost was reduced more with a higher verbatim reuse rate.
The breakeven point (1)	- ER16 (2011): The point that the costs of implementing reusable assets have been paid by the reuse benefits.	- ER16 reported that the case achieved the breakeven point only after a few implementations.
Cumulative development costs (1)	- ER24 (2011): Comparison between the cumulative development costs for products with and without Product Line Engineering(PLE).	- ER24 reported that PLE paid off the initial investment for this project by restricting core asset creation to a limited number of features.
Return on investment (1)	- ER27 (1998): For this application, a 1% improvement in the level of reuse achieved meant a saving in excess of £100,000.	- ER27 reported that software reuse could provide a convincing return on investment.

Table 10
Faster time-to-market metrics, definitions and results.

Metrics (number of primary studies used)	Definition from primary studies	Results
Delivery/Development time (4)	- ER2 (1994), ER19 (1995), QNL20 (2015), ER24 (2011): Time takes to deliver/develop the system/product/artifact.	- ER2, ER19, QNL20, and ER24 reported that development with software reuse took less time to market than development without software reuse.

of the operations/parameters/members/methods. The same goes for maintainability, that the overall effort used to fix or isolate the errors is reduced in reusable assets (QN3 [12], QN5 [23], ER16 [5]), but the proportion of errors requiring a longer fix is more in reusable assets (QN3 [12]). Ten primary studies (ER2 [7], QN7 [41], QN12 [15], QN15 [21], QN25 [45], ER19 [8], QN10 [4], QN26 [46], QN28 [47]) reported that software reuse increased apparent/gross productivity. This result is expected since software reuse reduces the effort of writing the code from scratch. Three primary studies (QN4 [2], QN10 [4], QN15 [21]) also reported that following a reuse approach allows developers to develop new assets faster than those without reuse. However, Frakes and Succi (QN7 [41]) observed a contradictory finding when comparing productivity with External Reuse Frequency (total number of references to reused external components/total number of references to reused both internal and external components) and External Reuse Level (number of external reusable assets that are used more than the maximum number of uses/total number of external and internal assets). Further investigation is needed for productivity, External Reuse Frequency, and External Reuse Level. Zhang and Jarzabek (QN11 [3]) and Beyer et al. (ER17 [9]) reported that software reuse benefits performance since it takes less space (ROM), while Deniz and Bilgen (QN15 [21]) found that software reuse causes lower performance since it takes longer to process (CPU usage, number of cycles, time delays). Slyngstad et al. (QL14 [16]) claimed that software reuse would benefit standardized architecture. However, when Kolb et al. (QN23 [44]) measured the conformance in architecture, the results showed that the conformance decreased. Further investigation into software reuse's impact on architecture is needed.

RQ3 Key findings: The key findings about the measurements for software reuse costs and benefits from the included 30 primary studies are as follows.

- Better quality and improved productivity are the most evaluated software reuse benefits in our sample.

Table 11
Other benefits, metrics, definitions and results.

Other benefits	Metrics (number of primary studies used)	Definition from primary studies	Results
Increased performance	Running time and memory usage (1)	- QN11 (2005): The authors measured the running time with the memory monitor and profiler turned on in the Wireless Toolkit.	- QN11 reported that the memory decreased, and the running time got faster after software reuse.
	Memory usage (1)	- ER17 (2008): The usage of ROM (i.e., the size of the binary program code).	- ER17 reported that development with systematic reuse had a lower usage percentage than development with ad hoc reuse.
Improved learning	Proxy of programmer's learning (1)	- QN4 (2002): The cumulated size of developed software (Object Oriented Function Points) up to the total number of developed applications.	- QN4 reported that the learning rate in development with reuse is more than the learning rate in development without reuse.
Prolonged lifecycle of reused product (game)	Grid Analysis (daily activity levels in the Game per 1 square km grid) (1)	- QN29 (2021): Average Actions per grid and day, Pre-new game release, Average Actions per grid and day, Post-new game release, Average number of old game GIP per grid, Average number of new game GIP per grid.	- QN29 reported that developing a new game with reusable components from the old game results in an increase in the number of active players in the old game.

- Most identified metrics used to evaluate software reuse benefits in better quality are defect-based (six out of 12), such as defect density.
- All identified metrics used to evaluate software reuse benefits in improved productivity are based on the development effort (six out of eight), such as apparent productivity.
- The identified metrics used to evaluate software reuse benefits in better maintainability (six) are based on maintenance effort (four out of six) and the amount of changed code (two out of six).
- The identified metrics used to evaluate software reuse benefits in cost-saving are about monetizing the return on investment (four out of six) and calculating the point when the costs of implementing reusable assets have been paid by reusing them (two out of six). However, the definition of these metrics is abstract compared to other impact measurements, like quality.
- The identified metrics used to evaluate software reuse benefits in faster time-to-market are about delivery time or development time.
- Software reuse may affect attributes in various ways using different metrics, e.g., software reuse improves the overall quality regarding defect density but may result in more defects that are severe or take more time to fix.
- All identified software reuse costs and benefits were measured, except standardized architecture (benefit) and increased learning curve (cost).

7. Strength of the evidence on software reuse costs and benefits (RQ4)

This section presents the strength of reported evidence regarding study types, software reuse costs and benefits, and quality assessment results.

7.1. Study type and quality assessment

We assessed the quality of the primary studies using the modified quality assessment checklist (as described in Section 3.4). We used nine out of 11 criteria from our quality assessment checklist to evaluate the primary studies' quality. We removed the first two criteria since they were used to screen out irrelevant papers. In addition, we categorized the primary studies into three study types: experience reports (ER), quantitative studies (QN), and qualitative studies (QL). To account for the variations in the study types, we customized the quality assessment criteria for each study type.

- ER: We removed the research design and reflexivity criteria for experience reports since experience reports are often written by practitioners who share their experiences or observations in a particular context and may not always follow the same formal research design and methodology required for a research study. In addition, the reflexivity criterion (relationship between researchers and participants) may not be necessary for experience reports since practitioners wrote them. Therefore, the maximum score for the experience report is seven.
- QL: We removed the criteria of the control group for qualitative study since qualitative studies only gather practitioners' perceptions about a specific project in a particular context, which does not require the control group. Therefore, the maximum score for the qualitative study is eight.
- QN: All criteria are applicable for quantitative studies, and the maximum score for quantitative study is nine.

We categorize the selected primary studies into three levels of quality.

- High-quality studies: the total score of the study is greater than 80% of the maximum scores (ER > 5.6, QL > 6.4, QN > 7.2).
- Moderate-quality studies: the total score of the study is less or equal to 80% of the maximum scores but greater than 50% of the maximum scores ($3.5 < ER \leq 5.6$, $4.0 < QL \leq 6.4$, $4.5 < QN \leq 7.2$).
- Low-quality studies: the total score of the study is less or equal to 50% of the maximum scores ($ER \leq 3.5$, $QL \leq 4.0$, $QN \leq 4.5$).

The primary studies have four high-quality primary studies, 20 moderate-quality primary studies, and six low-quality primary studies[7] (see Table 13). All experience reports fall under moderate quality due to a lack of rigorous data collection and analysis methods reporting.

7.2. Relating quality assessment results with software reuse results

We also mapped the quality assessment results with the study findings — software reuse costs and benefits. Fig. 6 depicts the primary studies' quality of the identified software reuse costs and benefits. All four high-quality primary studies reported better quality as one of the software reuse benefits. Furthermore, the overall quality of the primary studies that reported better quality benefits is higher than those that reported improved productivity benefits. Most of the primary studies that reported better quality (14 out of 20) and improved productivity (15 out of 20) benefits are of moderate quality. Primary studies that reported cost-saving and faster time-to-market benefits are of moderate quality, except ER27. Primary studies reported better maintainability

[7] The detailed score of each primary study is available at https://zenodo.org/records/10649264.

Table 12
Costs, metrics, definitions, and results.

Costs	Metrics (number of primary studies used)	Definition from primary studies	Results
Reduced quality	Error source & error type (1)	- QN3 (1997): Error sources: requirement, functional specification, design, code, or previous change. Type of errors: procedural error – a computational or a logic error; interface error – an internal or external interface error; data error – an initialization or a data value error.	- QN3 reported that modified components have increased design errors, new components have increased requirements and specification errors, and verbatim reused components have increased errors due to previous changes. In addition, modified components have more interface errors.
	Defect density per defect type/defect severity/impact (1)	- QN21 (2009): Defect density — number of defects/Non-commented Source Lines of Code (NSLOC). Defect types: Function, Assignment, Interface, Algorithm, Data, GUI, Relationship, Checking, Documentation, Timing/Serialization. Defect severity: Critical, High, Medium, Low, Very low. Defects based on impact attributes of Orthogonal Defect Classification (ODC): installability, integrity/security, performance, maintenance, serviceability, migration, documentation, usability, standards, reliability, and capability.	- QN21 reported that the defect types with the highest critical defect densities of the reused asset are different from those of the applications that are reusing it.
	Complexity (1)	- QN11 (2005): Complexity of the design: Maximum Size of Operation (MSOO); Maximum Number Of Levels (MNOL); Maximum Number Of Parameters (MNOP); Number Of Members (NOM); Number Of Remote Methods (NORM). Polymorphism: Number Of Overridden Methods (NOOM).	- QN11 reported that the complexity was increased slightly after using a reuse technique.
Reduced maintainability	Error isolation and completion effort (1)	- QN3 (1997): The authors categorized the effort to isolate and complete each error correction into three categories: less than one hour, one hour to one day, one to three days, and more than three days.	- QN3 reported that reused verbatim components had the highest percentage of errors requiring more than one day to complete an error correction, and the new components had the lowest percentage, while the modified components fell in between.
	Complexity and structuredness (1)	- QN23 (2006): Not reported	- QN23 reported that after reusing from the initially defined product line framework, the product showed a negative impact on maintainability and reusability compared to the initially defined product line framework.
Performance decay	Number of cycles (NoCycles) (1)	- QN15 (2014): The number of cycles (NoCycles) for the modules to receive the command from the system and send the corresponding data.	- QN15 reported that the number of cycles increases with increasing reuse.
	Time delay(ms) (1)	- QN15 (2014): The delay from receiving the track data from the system in three reuse scenarios (before software product line, software product line with pull strategy, software product line with push strategy).	- QN15 reported that the delay from receiving the track data was longer when the interface was changed to a more reusable and abstract level.
	CPU usage (1)	- QN15 (2014): The CPU usage during three scenarios (before SPL, pull strategy, push strategy).	- QN15 reported that the CPU usage increased when the interface was changed to a more reusable and abstract level.
Reduced productivity	Productivity (size of code per development effort) (1)	- QN7 (2001): The number of Non-Commentary Source Lines produced per person day (NCSL/Effort).	- QN7 reported that one case showed a negative relationship between External Reuse Frequency (ERF) and NSCL/Effort. Another case showed a marginally negative relationship between the External Reuse Level (ERL) and NSCL/Effort.
	Development effort per module (1)	- QN7 (2001): Effort in person days spent per module (effort/module).	- QN7 reported that one case showed a positive relationship between External Reuse Frequency and effort/module.
Low conformance in architecture	Conformance (1)	- QN23 (2006): Conformance is expressed in terms of percentage and calculated by the number of convergences divided by the total number of dependencies (i.e., divergences plus convergences).	- QN23: The proposed product line architecture is not fully conformant, and the product with the product line approach has the lowest conformance.

benefits are most of moderate quality. All identified software reuse benefits were reported by one or more than one primary study of moderate quality. As for costs, generally, primary studies that reported software reuse costs have lower quality than those reported software reuse benefits. Reduced quality, reduced maintainability, and reduced productivity are the costs that were at least reported by one moderate-quality primary study. Costs — performance decay, low conformance in architecture, and increased learning curve, were identified only by low-quality primary studies.

7.3. Relating quality assessment results with reuse-specific characteristics and software reuse results

We also mapped the quality assessment results with reuse-specific characteristics and the identified software reuse results.

The results show that high-quality studies are quantitative and only investigate systematic reuse. The high-quality studies are diverse in reuse approaches (see Table 13) and investigated verbatim or verbatim and modified reuse types with software reuse benefits in better quality, better maintainability, and improved productivity.

The moderate-quality studies investigated all identified reuse approaches and various reuse types and reuse methods. These studies have identified all software reuse benefits except increased performance. Two moderate-quality studies mentioned software reuse costs in reduced quality, reduced maintainability, and reduced productivity when performing verbatim reuse and ad hoc or systematic reuse.

Most of the software reuse costs were identified through low-quality studies. The low-quality studies often fail to report one or more than one reuse-specific characteristic.

Table 13
Mapping between quality assessment and software reuse results.

Study quality	Reuse approaches	Reuse type	Reuse method	Primary studies	Software reuse benefits	Software reuse costs
High-quality	SPL	Verbatim	Systematic reuse	QN13	Better quality, better maintainability	
	Reuse from repository	Verbatim and modified	Systematic reuse	QN10	Better quality, improved productivity	
	Compositional reuse	Not reported	Systematic reuse	QN1	Better quality	
	Framework reuse	Verbatim	Systematic reuse	QN21	Better quality	Reduced quality
	Template reuse	Verbatim and modified	Systematic reuse	QN10	Better quality, improved productivity	
Moderate-quality	SPL	Verbatim	Systematic reuse	QN3, QN9	Better quality, better maintainability	Reduced quality, reduced maintainability
		Verbatim and modified	Systematic reuse	ER8, ER19	Better quality, improved productivity, cost-saving	
		Modified	Not reported	QN29	Prolonged lifecycle of reused product (game)	
		Not reported	Systematic reuse	ER18, ER24	Better quality, improved productivity, cost-saving, faster time-to-market	
			Ad hoc reuse	ER17	Better quality, increased performance, improved productivity	
			Not reported	ER16	Better quality, better maintainability, improved productivity, cost-saving	
	Reuse from repository	Verbatim	Systematic reuse	ER6, QNL20	Improved productivity, cost-saving, faster time-to-market	
			Controlled reuse	QN12	Better quality, improved productivity, cost-saving	
		Verbatim and modified	Controlled	QN25	Better quality, improved productivity, cost-saving	
		Not reported	Systematic reuse	QN28	Improved productivity	
	Component reuse	Verbatim and modified	Not reported	QL14	Better quality, standardized architecture, improved productivity, cost-saving	
	Compositional reuse	Verbatim	Ad hoc reuse	QN7	Better quality, improved productivity	Reduced productivity
	Framework reuse	Verbatim and modified	Systematic reuse	QN4	Better quality, improved productivity, improved learning	
	Template reuse	Modified	Not reported	QN26	Better quality, improved productivity	
	Architecture reuse and context-dependent module reuse	Verbatim and modified	Systematic reuse	QN5	Better quality, better maintainability, improved productivity	
	Not reported	Verbatim and modified	Systematic reuse	ER2	Better quality, improved productivity, cost-saving, faster time-to-market	
Low-quality	SPL	Modified	Not reported	QN11	Better quality, better maintainability, increased performance, improved productivity	Reduced quality
		Not reported	Systematic reuse	ER22, QN23	Improved productivity	Reduced maintainability, low conformance in architecture, increased costs
			Not reported	QN15	Better quality, improved productivity	Performance decay
	Engine reuse	Not reported	Not reported	QN15	Better quality	Performance decay
	Architecture reuse and context-dependent module reuse	Modified	Not reported	QL30	Improved productivity	Increased learning curve
	Not reported	Verbatim and modified	Systematic reuse	ER27	cost-saving	

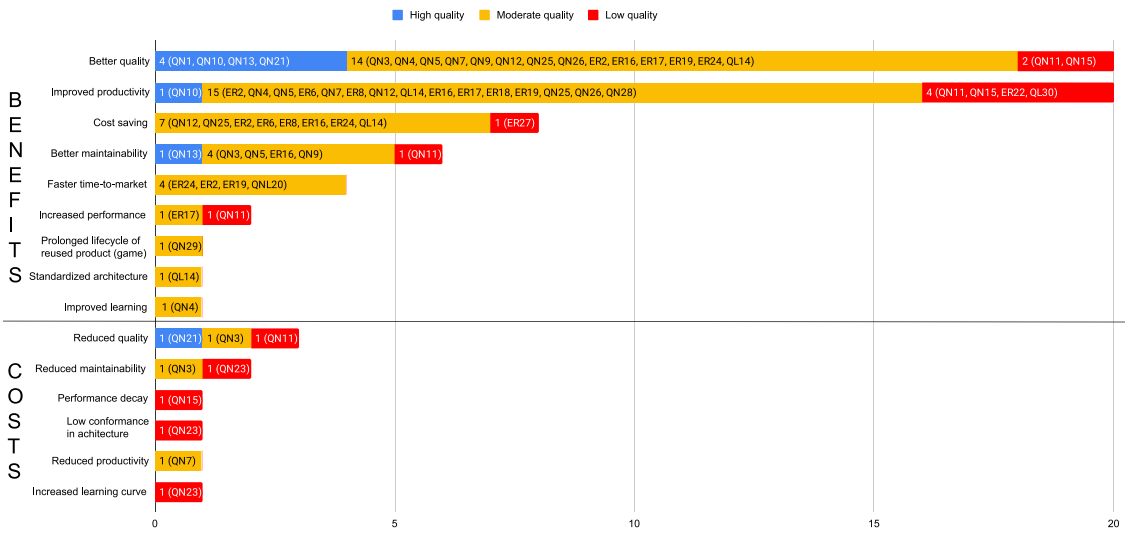

Fig. 6. Software reuse costs, benefits, and quality assessment results.

7.4. Validity threats of the primary studies

Validity threats demonstrate how researchers address their study limitations, which also presents the strength of evidence. Only 11 out of 20 primary studies (excluding ten experience reports) - two qualitative and nine quantitative studies, reported validity threats. Seven of these 11 primary studies have discussed the validity threats extensively, while the other four only mentioned some threats without detailed descriptions. We discuss the reported threats to validity in four types as follows.

Construct validity is concerned with whether the selected metrics or questionnaire instruments reflect the intended construct. For quantitative studies, researchers discuss construct validity from four perspectives: study design, justification of the selected metrics and data, confounding factors, and mitigating solutions. Seven primary studies discussed construct validity. Morisio et al. (QN4 [2]) justified how the selected metrics measure the target attributes — productivity, quality, and learning. In addition, they also described the problem of measuring developers' learning and provided mitigate solutions — using the learning metrics used in manufacturing. Gupta et al. (QN21 [43]) justified that the selected data helped answer the research questions, and they triangulated the results with interviews. Goldin and Berry (QNL20 [53]) also mentioned interview triangulation to address the understanding of the measurements and their purpose. Deniz and Bilgen (QN15 [21]) discussed the reliability of the data in terms of whether it is measured and collected according to the theoretical constructs. Mihale-Wilson et al. (QN29 [48]) lacked revenue and cost information for the profit impact of reusing components in location-based games. Mohagheghi et al. (QN9 [13]) and Mohagheghi and Conradi (QN13 [42]) selected the metrics based on the lesson learned from existing literature. As for qualitative studies, researchers discuss construct validity in terms of the construction of the questionnaire.

Internal validity reflects the causal relationship the researchers are investigating is not influenced by other factors. In quantitative studies, the researchers discuss mainly three types of confounding factors: missing, inconsistent, and wrong data; the causality between the input and the responses; and confounding factors to the investigated factor. 11 primary studies reported internal validity. Mohagheghi and Conradi (QN13 [42]), Mohagheghi et al. (QN9 [13]), and Gupta et al. (QN21 [43]) discussed the missing and incomplete data threats due to the reporting process in the company, and provided mitigation and its justifications. Succi et al. (QN1 [1]) also mentioned that the data collection relies much on the company's standard procedure. Morisio et al. (QN4 [2]) and Goldin and Berry (QNL20 [53]) have mentioned the causality in design and the assumption causality. Mao et al. (QN26 [46]) reported that they could not afford to have two teams with similar experience levels develop the same project with and without following software reuse practices due to limited resources. To address this, the authors asked the same team to develop projects with similar complexity with and without following software reuse practices. Four studies mentioned the confounding factors: programming language, reusable assets have more work from experienced developers or testers, different functionality and constraints between reused and non-reused components (QN9 [13], QN13 [42]), knowledge and skill improvement of developers [46], and enthusiasm of trying the new approach (QNL20 [53]). The internal threats in qualitative studies are related to whether participants answered the questionnaire truthfully.

External validity refers to what extent the results can be generalized to another context. Ten primary studies have reported external validity or generalizability. All ten primary studies discussed a common limitation: they noted that the results were limited in the investigated company context. Considering the specific context and sampling size of the studied object or participants, five of these ten studies (QL14 [16], QL30 [52], QN4 [2], QN21 [43], QN15 [21]) mentioned the results cannot be generalized to other companies, while the other five (QN1 [1], QN9 [13], QN13 [42], QN29 [48], QNL20 [53]) claimed that the results could be applicable for similar projects with similar context.

Conclusion validity refers to whether the results are statistically significant or reliable. Seven primary studies discussed the conclusion validity. Mohaghehi et al. (QN9 [13]) and Gupta et al. (QN21 [43]) discussed the factors that may influence the results. Mohagheghi et al. (QN9 [13]) acknowledged that different ways of handling user interface in reused and non-reused components could be internal validity in measuring the reuse impact. In addition, they also acknowledged the limitations of the defects coverage in the collected Trouble Reports. Gupta et al. (QN21 [43]) and Mohagheghi et al. (QN9 [13]) mentioned that the developers' experiences and skill levels would not be a threat since the assets under study are developed within the same development unit. Slyngstad et al. (QL14 [16]), Succi et al. (QN1 [1]), and Moriso et al. (QN4 [2]) discussed the statistic significance in regards to the population and confirmed the value of the results. Goldin and Berry (QNL20 [53]) and Deniz and Bilgen (QN15 [21]) have discussed the study's replicability when conducted by different researchers or analysts and claimed that the results would be similar.

RQ4 Key findings: We customized the quality assessment checklist developed by Dybå and Dingsøyr [32] and assessed the quality of the included 30 primary studies. The key findings about the quality of the included primary studies are as follows.

- Most of the primary studies (20 out of 30) are of moderate quality. Four primary studies are of high quality, and six are of low quality.
- Better quality and improved productivity are reported in all quality studies. All four high-quality studies reported that software reuse benefits quality and one reported that it benefits productivity. The results showed strong evidence behind the claims that software reuse improves quality and productivity. Moreover, the studies' rigor behind the claim that software reuse improves quality is better than productivity.
- Our validity threats results showed that researchers need to pay more attention to reporting validity threats. We found a relatively low number of included quantitative and qualitative studies (11 out of 20, excluding ten experience reports) reported validity threats, and not all of the 11 primary studies reported validity thoroughly.
- Experience reports often get lower scores in data collection and analysis criteria, which require reporting formal data collection and analysis methods.
- Quantitative studies have lower scores in research design and reflexivity, which require discussing and justifying why the authors decided which research method to use and whether their roles influence sample recruitment and data selection.

8. Discussion

This section discusses the difference between our study findings and other secondary studies and the study implications for researchers and practitioners.

8.1. Comparison with existing secondary studies on software reuse costs and benefits

We mainly discuss our findings with secondary studies from Mohagheghi and Conradi [17] and Barros et al. [20]. In the primary studies, the most frequently adopted reuse approaches are SPL and reuse from repository. The results of Mohagheghi and Conradi [17] showed more studies are reusing libraries or repositories, while Barros et al. [20] identified component-based development and SPL are the most practiced reuse approaches. In addition to the reuse approaches identified by Mohagheghi and Conradi [17] and Barros et al. [20], we found a new reuse approach: engine reuse [21], which refers to reuse an automated process that generates or finds/selects the components with the needed input information. The emerging engine reuse may result from the evolving software engineering technology. With time passing by, more reuse approaches may arise, such as microservices reuse [55] and InnerSource reuse [56]. Regarding reuse types, our results showed that verbatim reuse is applied more than modified reuse, and ten primary studies (43%) used a combination of verbatim and modified reuse. Similar findings were observed in Mohagheghi and Conradi [17], where five studies used verbatim reuse, and six used a combination of verbatim and modified reuse in their included 11 primary studies. As for reuse methods, like Mohagheghi and Conradi [17] and Barros et al. [20], we found that the included industrial cases investigated systematic reuse more than the other reuse methods — 20 reuse cases followed systematic reuse (74%), while four followed ad hoc reuse (15%) and three followed controlled reuse (11%). In addition, the primary studies report 18 domain reuse cases, which is more than infrastructure (five reuse cases) and architecture reuse (one reuse case), aligning with the results of Mohagheghi and Conradi [17].

We also investigated the comparison scheme that the primary studies used to evaluate software reuse costs and benefits. The most common reuse comparisons are three types: comparing the original project developed from scratch and the projects reused the original project [8,13,43,44,48,50,51,53], comparing the assets that follow or do not follow a reuse approach [1–6], and comparing the assets with different reuse rates [15,21,41,45,47]. These three comparison schemes are similar to the ones Mohagheghi and Conradi [17] extracted — whether development happens with or without systematic reuse and reuse rate. Few studies [12,23] investigated the reuse impact based on modification rate, which is also mentioned by Mohagheghi and Conradi [17]. In addition to what Mohagheghi and Conradi [17] reported, the primary studies also compared the reuse impacts based on the reuse methods (whether following systematic reuse) [9,10,14] and between the reused and new assets in the same project [7,13,42].

The most investigated software reuse benefits in the primary studies are better product quality and improved productivity, which align with what Mohagheghi and Conradi [17] and Barros et al. [20] reported. We found one additional software reuse benefit: prolonged lifecycle of reused product (game) [48]. Like Mohagheghi and Conradi [17] and Barros et al. [20], we also noted that fewer studies reported software reuse costs: two out of 11 primary studies in Mohagheghi and Conradi [17], two out of 49 primary studies in Barros et al. [20] and seven out of 30 primary studies in our systematic literature review. In addition, we also assessed the study quality to understand the strength of the evidence behind the identified software reuse costs and benefits. The results showed strong evidence behind the claims that software reuse improves quality and productivity.

Regarding reuse methods, our results showed that more primary studies investigated systematic reuse and its benefits (19 primary studies, 76%) over controlled reuse (three primary studies, 12%) and ad hoc reuse (three primary studies, 12%), which aligns with the results of Barros et al. [20]. As for reuse approaches, our results showed that all reuse approaches benefit the companies in quality, and SPL and reuse from repository are most likely to benefit companies. Barros et al. [20] found that component-based development, SPL, and model-based development can all benefit product quality, productivity, development cost, and development time, whereas component-based development has more evidence to benefit companies.

The primary studies have used different metrics for measuring software reuse costs and benefits, especially for quality and productivity. In the primary studies, defect density, complexity, number of faults/defects/bugs, and the effort to fix them are the most common metrics used to measure software quality, which is in line with what Mohagheghi and Conradi [17] reported. In addition to these metrics, some primary studies have instead used software coupling [3,21], the developers' [41] and customers' [1,9] views on the product quality to measure quality. Like Mohagheghi and Conradi [17], the extracted productivity metrics are mostly used to measure the effort spent developing and designing reusable assets. Moreover, we also found that the primary studies measure the testing effort [9] and the amount of produced assets [23]. The metrics used to measure maintainability are the same as what Mohagheghi and Conradi [17] reported. Compared to Mohagheghi et al. [17], we also reported metrics used to measure software reuse effects other than software quality, productivity, and economics, e.g., learning and architecture conformance.

We extracted six monetized cost-saving metrics from primary studies. However, we found that those monetized cost-saving metrics lack clear definitions (see Section 6). Barros et al. [20] noted similar findings that their included primary studies did not monetize the costs of development time reduction, quality increase, and faster time-to-market. Quantifying the reuse benefits monetarily is important, and it is a more intuitive way for companies in decision-making.

8.2. Study implication for researchers and practitioners

Based on our results, this subsection presents the possible implications for practitioners and researchers.

- Quantifying the reuse benefits monetarily is important, providing a more intuitive way for companies to make decisions. As discussed in Section 8.1, the extracted six monetized cost-saving

metrics lack clear definitions compared to other extracted quality or productivity related metrics. Clear definitions for monetized metrics are needed so that the researchers and practitioners can understand how the software reuse costs and benefits are measured.

- Most primary studies are about software reuse benefits, with only a few focusing on costs. There is a need to investigate more into software reuse costs, especially the tradeoff between software reuse costs and benefits (e.g., the tradeoff between quality increases regarding overall defect density and quality decrease regarding the number of severe defects or the defects that take more time to fix).
- Mäkitalo et al. [57] found that opportunistic reuse is getting increasingly popular, especially reusing from open source software. It will be interesting to investigate the differences in reusing external and internal reusable assets.
- Some identified costs and benefits from qualitative studies need to be validated by quantitative data, such as increased learning curve and standardized architecture.
- Researchers should improve their quality of reporting. Researchers need to report all necessary characteristics and validity threats to help understand the context of the investigated case, which further contributes to the generalizability of the study findings. According to the quality assessment results, the researchers can improve the quality of their reports by justifying the reason for their selected methods, improving sampling, and reflecting more on their role implications to the study.
- The evolution of technologies and methods, such as microservices and InnerSource, may impact software reuse differently. Microservices is one type of reusable unit, and it benefits software reuse [55,58]. InnerSource benefits software reuse [56,59], and its shared space is closely related to reuse from repository. However, the studies above only mentioned microservices and InnerSource benefits software reuse. Qualitative and quantitative evidence about how microservices and InnerSource influence software reuse has not yet been investigated. Our systematic literature review did not find studies reporting software reuse costs and benefits from InnerSource and microservice. The reasons for not finding such studies are as follows: (1.) The identified systematic secondary studies focus mainly on traditional software reuse practices, and (2.) Most of the InnerSource and microservice reuse related studies only mentioned that they improve software reuse, but it is unclear which aspects and to what extent they improve software reuse. (3) Microservices and InnerSource are relatively new approaches. Chen et al. [11] conducted an exploratory case study to address this gap and investigated the impact of microservices and InnerSource on software reuse from the practitioners' views. However, more software reuse investigations on InnerSource and microservices are needed.
- Our review provides practitioners evidence that software reuse benefits companies, especially in improving software quality and productivity. Our sample showed that development with reuse reported all of the identified benefits with only a few costs. Developing reusable assets has upfront costs [7,14], which will be paid by reusing them to develop other products [5,52]. In addition, Baldassarre et al. [4] found that a systematic way of reuse will improve the productivity in development for reuse compared to conventional development. Systematic reuse is the most likely optimized option compared to other software reuse practices. Reusing as-is - verbatim reuse has more benefits than modified reuse. Modified reuse is needed when the assets cannot be reused as is. However, extensive modification is not recommended due to its higher risk in quality issues. No matter what reuse approaches the companies use, the discoverability of the reusable assets is important. Investigating how companies ensure the discoverability of reusable assets will be an interesting topic.

9. Conclusion and future work

In this study, we followed the guidelines from Kitchenham et al. [28] and conducted a systematic literature review on software reuse characteristics, costs, and benefits. In total, we included 30 primary studies.

Within these 30 primary studies, 29 reported software reuse benefits, and seven reported software reuse costs. We identified nine software reuse benefits and six software reuse costs. The most reported software reuse benefits are better quality and improved productivity. We cannot draw any conclusions on software reuse costs because of their limited number of studies. We identified eight reuse approaches, of which SPL and reuse from repository were the most reported. Almost all reuse approaches resulted in better quality and improved productivity. Verbatim reuse and systematic reuse were used the most and benefited the companies over the other reuse types and methods.

We extracted all the metrics used to evaluate software reuse costs and benefits and found that the primary studies adopted multiple metrics to evaluate quality and productivity. More metrics are needed to validate other identified software reuse costs and benefits. Software reuse impacts on one attribute may vary according to the metrics from different aspects. For example, the primary studies showed that software reuse helped decrease the overall software defect density, but certain defect types and critical defects have increased.

We also examined the quality of the reviewed primary studies using a quality assessment checklist and categorized them into different study types. The quality assessment results showed that most primary studies are of moderate quality. All types of quality studies reported that software reuse benefits quality and productivity, indicating there is strong evidence behind the claims that software reuse benefits product quality and productivity.

We identified some research gaps for future research as follows:

- To investigate what software reuse costs and benefits the microservice and InnerSource can bring to the companies.
- To provide the quantitative evidence for the software reuse benefits that were not measured: standardized architecture and increased learning curve.
- To provide more empirical evidence on the less reported software reuse costs and benefits.
- To investigate how to monetize the software reuse costs.
- To investigate which context information is important to be reported in software reuse cases.

CRediT authorship contribution statement

Xingru Chen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing, Project administration. **Muhammad Usman:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Validation, Visualization, Writing – review & editing. **Deepika Badampudi:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

https://zenodo.org/records/10649264.

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